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RESEARCH ARTICLE

Facial expression recognition on partially occluded faces using component based ensemble stacked CNN

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Abstract
Facial Expression Recognition (FER) is the basis for many applications including human-computer interaction and surveillance. While developing such applications, it is imperative to understand human emotions for better interaction with machines. Among many FER models developed so far, Ensemble Stacked Convolution Neural Networks (ES-CNN) showed an empirical impact in improving the performance of FER on static images. However, the existing ES-CNN based FER models trained with features extracted from the entire face, are unable to address the issues of ambient parameters such as pose, illumination, occlusions. To mitigate the problem of reduced performance of ES-CNN on partially occluded faces, a Component based ES-CNN (CES-CNN) is proposed. CES-CNN applies ES-CNN on action units of individual face components such as eyes, eyebrows, nose, cheek, mouth, and glabella as one subnet of the network. Max-Voting based ensemble classifier is used to ensemble the decisions of the subnets in order to obtain the optimized recognition accuracy. The proposed CES-CNN is validated by conducting experiments on benchmark datasets and the performance is compared with the state-of-the-art models. It is observed from the experimental results that the proposed model has a significant enhancement in the recognition accuracy compared to the existing models.

Keywords Facial expression recognition · Action units · Face components · Partially occluded faces · Ensemble stacked CNN

Introduction

The facial expression represents several facial emotions is a non-verbal communication tool that humans use in their day-to-day life. This is a mechanism that shows the real and actual feelings of humans as reactions in communication. Facial Expression Recognition (FER) is an inevitable aspect of identifying such emotions of humans from their facial expressions. Automation of FER is crucial in developing Human-Computer Interaction (HCI) machines which aid to improve the quality of life via many diverse applications such as image re-ranking (Yu et al. 2012, 2014a, b, 2016), health monitoring systems (Muhammad et al. 2017), interrogations (Hsu et al. 2017), e-commerce, facial attractiveness (Zhao et al. 2020), etc. The FER models train the machines towards understanding human emotions for better HCI performance using machine learning or deep learning techniques. There exist many FER models that developed based on human emotions extracted from static face images (Liu et al. 2014; Shan et al. 2009; Chao et al. 2015; Danisman et al. 2013; Fu et al. 2019). In general, there are two types of FER models: feature based models and deep learning based models. The conventional feature based models typically involve three stages (Jain and Li 2011): face detection, feature extraction, and classification. The challenging tasks

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among these stages are face detection, which involves effectively locating the face regions, and, feature extraction, which extracts the significant facial features that enhance the classification accuracy.

In general, the features of different expressions are diverse depending on the Action Units (AUs) such as raising eyebrows or upper lids, nose wrinkles, cheek puffers, tightening or widening lip parts, squinting eyes (as depicted in Fig. 1). These AUs contribute a major portion in assisting FER systems to identify facial expressions. Principal Component Analysis (PCA) (Jain and Li 2011), Linear Discriminant Analysis (LDA) with Independent Component Analysis (ICA) (Bellamkonda and Gopalan 2018a), Local Binary Patterns (LBP) (Ramachandran et al. 2015b; Bellamkonda and Gopalan 2018b), expression-specific LBP (Chao et al. 2015), Gabor wavelets features (Zhang et al. 1998) and Gabor wavelets with Local Directionality Patterns (LDP) (Shan et al. 2009) are some of the approaches used by the existing models to detect face regions (based on appearance) and to extract the features from the whole face regions. Some studies (Wang et al. 2013; Gopalan et al. 2018; Bellamkonda and Gopalan 2020; Avani et al. 2021) developed FER models by identifying the face regions based on geometric landmark positions. And finally, the third stage is a classification of face expressions into seven basic emotions (Ekman 1993) based on the extracted features. The classification was done by different machine learning techniques such as K-nearest neighbor (Bellamkonda and Gopalan 2020), Support Vector Machines (SVM) (Ouyang et al. 2015). Rather than extracting features from the face images, the deep learning based FER models considered fixed size patches of images in random order as input. The deep learning techniques such as Convolution Neural Networks (CNN) (Wang et al. 2013), Artificial Neural Networks (ANN) (Gopalan and Bellamkonda 2018), backpropagation neural networks (Ramachandran et al. 2015a, b), and deep belief networks (Ramachandran et al. 2013; Uddin et al. 2017; Kurup et al. 2019) have been used in deep learning based FER models to classify the face images into basic emotions.

The problem that existed in most FER models (feature based or deep learning based) is performing feature extraction based on the AUs of non-occluded facial images taken under controlled laboratory conditions (Wen et al. 2017). However, in the real world, the complications arise in recognizing facial expressions due to the presence of occlusions such as wearing sunglasses, scarves, caps, face masks, or having a beard and or any other obstruction that comes in between while capturing the images as shown in Fig. 2. These complications show a great impact on the performance of recognizing facial expressions. The existing FER models that were trained on non-occluded faces are not robust to recognize the facial expressions from occluded facial images. The other complications involved with the FER system are different pose changes, illuminations, geometric position invariances, unwanted noise such as hair or background in the facial image. To mitigate the problem of reduced performance in detecting facial expressions due to these distortion aspects, CES-CNN model is proposed to perform FER on individual action units to improve emotion recognition performance. The proposed model divides the entire face image into six parts (Yin et al. 2021) namely eyebrows, glabella, eyes, nose, cheeks, and mouth using the Viola-Jones face detection algorithm. The face components detected by the Viola-Jones algorithm are in different sizes. To maintain uniformity while training using CNN subnets, all the face components are rescaled to the same size. Further, Component based Ensemble Stacked Convolution Neural Networks (CES-CNN) is used to train each action unit to classify seven basic emotions (neutral, surprise, happy, sad, angry, fear, and disgust) which were identified as universal across different cultures by Ekman (1999a, b). A stack of six subnets is used where each subnet contains a CNN network to train the model with seven basic emotions from

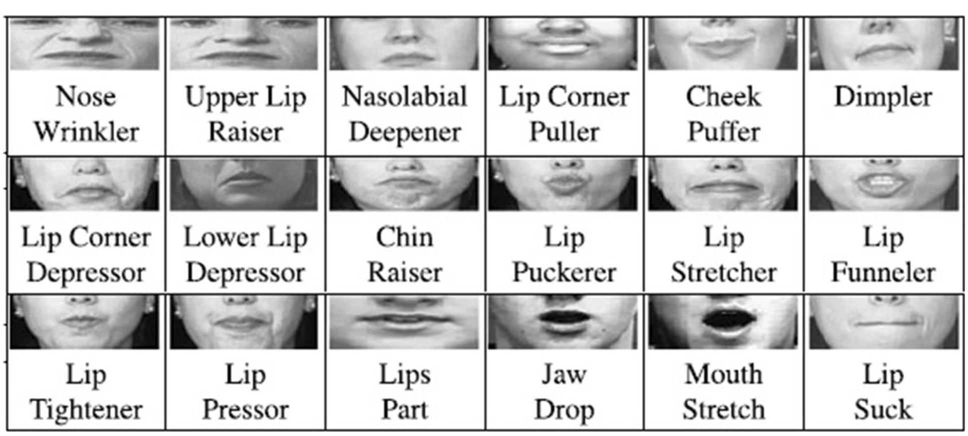


Fig. 1 Action Units of traditional FER system

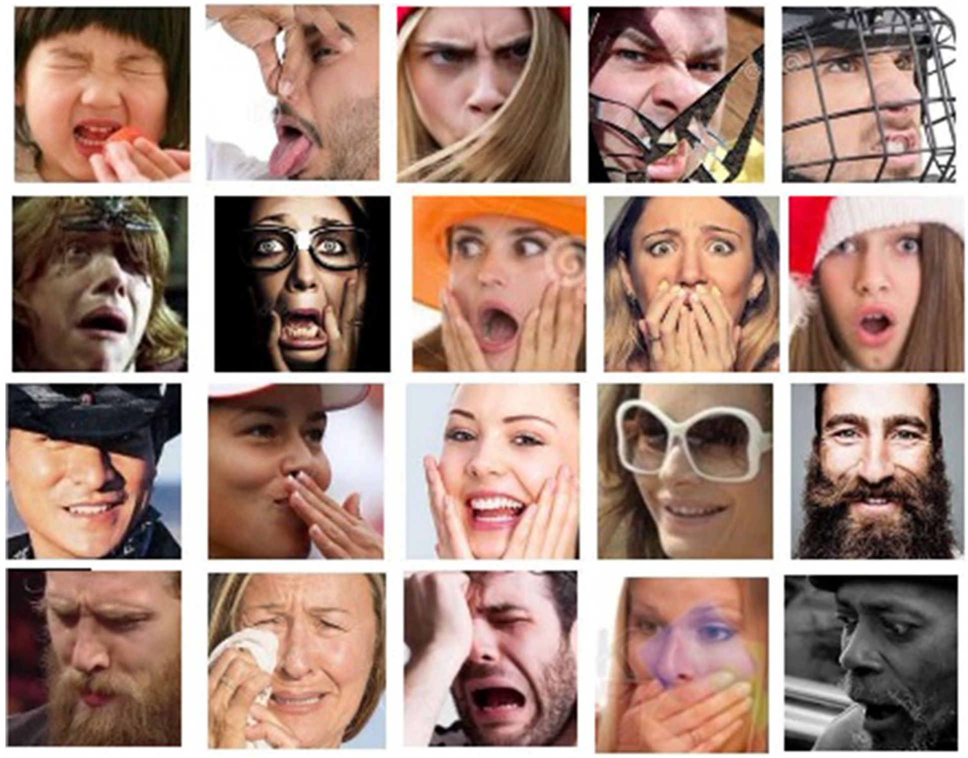


Fig. 2 Partially occluded facial images for different emotions

each face component. The decisions from each subnet of CNN are henceforth combined to produce an optimized result of emotion recognition using a Max-Voting classifier.

The salient contributions of this work are summarized as follows:

1. We propose a Component based Ensemble Stacked Convolution Neural Networks CES-CNN to recognize facial expressions from partially occluded faces. CES-CNN is robust architecture to apply on facial images taken under normal conditions or various distorted conditions.
2. The CES-CNN architecture can enhance the performance of FER without any ensemble pruning mechanisms such as feature selection or committee member selection.
3. Max-Voting based ensemble classifier used along with the CNN in the proposed model improves the recognition performance by defining spontaneous facial expressions which include different occlusions.

The rest of the paper is organized as follows: Sect. 2 reviews the existing works on FER from partially occluded faces and FER using ensemble stacked deep neural network approaches. Section 3 discusses the proposed FER model on partially occluded faces including face components detection using Viola-Jones algorithm, Component based Ensemble Stacked CNN (CES-CNN), and Max-Voting based ensemble classifier that used to produce the ensemble result of the CES-CNN. Section 4 presents the experimental results and comparison analysis with state-of-the-art models of occlusion aware FER followed with the concluding remarks in Sect. 5.

Literature review

In recent years, numerous FER models were developed among which some existing works are reviewed based on two aspects i.e the FER models related to mitigating the same issue (FER on partially occluded faces) and models developed using similar network design (ensemble deep learning networks).

Ensemble deep learning models for FER

Ensemble of deep learning model devises a set of deep learning networks in such a way that combining the decisions of an individual network can enhance the performance of the model. The ensemble approach can resolve most of the issues including the classification of an instance with unseen feature sets. Moreover, many ensemble deep learning models (Liu et al. 2016; Sun et al. 2019; Nguyen et al. 2019; Wang et al. 2021) proved that ensemble techniques improve FER accuracy. While the use of ConvNets (Lopes et al. 2017; Kim et al. 2018) with VGG (Visual

Geometry Group) as fine-tuning the images achieved high accuracy in FER problems among the models developed so far. Whereas, Sun et al. (2018) developed a FER model to extend the ensemble model to analyze the expressions from four different kinds of conditions in faces such as aligned frontal faces, faces in different poses, aligned unconstrained faces, and grouped unconstrained faces. They fabricated an embedded attention model as a second layer in the three layered architecture to determine regions of interest (ROI) (Lopes et al. 2017) from local features of faces obtained from the first layer. The local features of interested regions are then fused in the third layer to infer the facial emotion.

There exist some other FER models which developed ensemble deep learning networks in different ways to improve the performance of FER. Wen et al. (2015, 2017) proposed two approaches based on ensemble deep neural networks in application to FER. First approach ensemble convolutional echo state network to mitigate the problems such as vanishing gradients, over-training, the existence of differences in face images due to age and ethnicity, multiple parameters tuning, and dubious class labels that commonly occur in FER models. This model considers Convolutional Network (CN) to transform the face image with random parameters and architectures to produce diverse classification results. Echo State Network (ESN) was then used as the base classifier to ensemble the diverse results in order to produce the superior result of FER. The second approach ensembled the Convolutional Neural Networks (CNNs) with probability-based fusion. This approach considered CNNs with convolutional rectified linear layer as the first layer and multiple hidden maxout layers. The diversity in different CNNs was achieved by varying parameters and architectures to obtain different probabilities for each class as output. Then the different output probabilities were fused using the probability-based fusion method to produce the optimal FER result. (Shao and Cheng 2021) proposed an edge-aware feedback convolutional neural network (E-FCNN) which contains a three-stream super resolution network to effectively analyze the textual information of faces. E-FCNN is a network embedded with three branches: an edge-enhancement block, up-sampling branch, and SR(Super-Resolution) primary branch. A hierarchical strategy is used to extract the visual features from tiny images which are then forwarded to a feedback block along with fused results of the three branches for facial expression recognition.

Wang et al. (2018) proposed CNN Ensemble for single face images and concluded based on the experimental results on the FER2013 dataset that there is a significant improvement in the performance of FER accuracy. FER model by Xie and Hu (2018) has divided the CNN into two branches for the effective performance in which one branch has been used to extract the local features from image patches whereas the other branch has been used to extract the holistic features from the whole expression image. Dynamically Weighted Majority Voting (DWMV) based ensemble classifier is proposed in Zia et al. (2018) to infer the optimized result of FER. DWMV based ensemble classifier recognizes spontaneous facial expression which is included with different muscle movements in a single expression. Kim et al. (2016) developed a robust FER model hierarchical committee of deep convolutional neural networks. This model improved the FER accuracy by obtaining diverse decisions from the hierarchical deep CNNs trained with input normalization and random weight initialization. The hierarchical architecture of deep CNNs was constructed by the model based on the exponentially-weighted decision fusion. The FER model proposed in Wang et al. (2020) used multi-scale block local binary pattern uniform histogram (MB-LBPUH) to represent the holistic structural features from facial images. Then a novel feature representation is created for facial expressions by fusing MB-LBPUH features and HOG features. Parallelly, weights are assigned to the MB-LBPUH feature to avoid unbalancing in fusing the features. And then the principal component analysis is used for dimensionality reduction and support vector machine for classification.

Some FER models (Li and Wen 2018; Li et al. 2019a, b) also proposed different ensemble pruning algorithms such as MRMREP (Maximum Relevance and Minimum Redundancy-based Ensemble Pruning), GDEP (Graph-based Dynamic Ensemble Pruning), and RTCRelief-F respectively to mitigate the problem of redundant members in classifier pools of ensemble FER models by attaining the more compact subset of classifier pool which enhance the recognition accuracy. The proposed MRMREP, GDEP, and RTCRelief-F ensemble pruning algorithms select the classifiers based on the diversity as well as the performance. MRMREP is a maximum relevance and minimum redundancy-based ensemble pruning algorithm which considers the predictions of classifiers as features to the feature selection method to attain a more representative subset. GDEP is a graph-based dynamic ensemble pruning algorithm that measures the statistics of classifiers' behavior and builds the neighbor graph based on the similarity in classifiers' performance. RTCRelief-F is clustering based and ordering based ensemble pruning algorithm which clusters the classifiers and resets the fusion order based on classifiers' ability.

Ali et al. (2020) developed a multicultural FER model using an ensemble classifier approach. This model proposed a two-level feature extraction using Local Binary Patterns (LBP), Uniform LBP (ULBP) which attains facial features with many dimensions. Then the Principal Component Analysis (PCA) was used to reduce the dimensionality of the feature vector. Then the facial feature vectors of cross-cultural faces were analyzed with ensemble classifier of Binary Neural Networks (BNNs) as three levels: base-level, meta-level, and predictor to predict the presence of expression in the face. While the binary decision on face expression was done by the fusion of Naive Bayes (NB) with Bernoulli distribution (BD). Edge-based feature extraction method such as Histogram of Oriented Gradients is used in Nazir et al. (2018) along with embedded classifier to improve the FER performance. Minaee et al. (2021) used attention neural networks in an end-to-end framework for FER which focused on the important facial features in an image to improve the recognition performance.

FER models on partially occluded faces

The real-world applications of FER have to confront many challenges such as head pose variations, occlusion, lightening conditions, and poor image quality that hinder the recognition performance. Many researches have been evolved towards improving the FER accuracy from the facial images of unconstrained environments. There are two categories of FER models that address the facial occlusions. One is holistic-based models which consider the still image as a whole while training the FER model and the other is part-based models which divide the face into several overlapped or non-overlapped sub-regions. The holistic-based models such as (Wright et al. 2008) used L1-norm regularization to enhance the robustness of FER, however, L1-norm regularization is not acceptable for non-facial occlusions. The FER model developed by Ranzanto et al. (2011) is another holistic-based approach that used generative networks to construct the entire face from the occluded face. However, the reliability of the model that used generative approaches changes based on the training data with various occlusion conditions. Liu et al. (2018) also proposed a holistic-based model of conditional convolutional neural network enhanced random forest (CoNERF) to mitigate the issues with various distortion aspects such as occlusions and illuminations. This model extracted robust deep salient features from saliency-guided facial patches in order to mitigate the distortion issues. Then FER is conducted using CoNERF which is an enhanced decision tree model with representation learning that split the nodes using a neurally connected split function (NCSF) and conditional probabilistic learning that recognizes emotions from faces with different perspectives.

Dong et al. (2019) proposed non-convex low-rank decomposition in the training phase of the FER model to produce an expression based dictionary. In the testing phase, an occlusion error model is proposed in order to represent the biased information of occluded faces based on the dictionary produced in the training phase. Liu et al. (2021) proposed a dynamic multi-channel metric learning network (DML-Net) to develop a FER model to handle poses and identity invariants in facial images. DML-Net is constructed with three parallel multi-channel convolutional networks to be aware of fused global and local features from different facial regions. Henceforth, joint embedded feature learning is used to identify identity-invariant and pose-aware expressions in an embedding space that contains fused region-based features. Whereas (Zhang et al. 2016) used principal component analysis network (PCA-Net) incorporated with convolutional neural networks (CNN) as PRP-CNN to develop a pose-aware and identity-invariant expressions from facial images. Poux et al. Jie and Yongsheng (2020) proposed an MVFE-LightNet (Multi-View Facial Expression Light Weight Network) to recognize expressions from faces with various poses. MTCNN is applied initially to detect and align faces from images and then performed normalization and augmentation to preprocess the facial images. Then MVFE-LightNet is applied to extract sub-space features of facial expressions with various poses. To handle the overfitting problem and reduce parameters, this model proposed a depthwise separable residual convolution module architecture.

Poux et al. (2021) proposed an expression based binary classifier which is developed based on the propagation properties of the facial movements. To mitigate the problem of occluded faces, this model stressed the importance of other unoccluded faces to grab an expression using binary classifiers. Then the outputs of the binary classifier for each expression are aggregated using a fusion process to produce improved recognition of facial expression. The FER models developed in (Martinez 2002; Zhong et al. 2012; Huang et al. 2012) are part-based which divided the facial images into uniform regions. The other part-based models (Li et al. 2017; Dapogny et al. 2018) divided the facial images into patches based on facial landmark positions. The part-based FER model developed by Zhang et al. (2014) divided the facial image into different parts and ignored the parts with occlusions while training the FER model. However, some part-part based FER models (Towner and Slater 2007; Deng et al. 2009; Mao et al. 2009; Lin et al. 2013) detected the occluded parts and compensated them using different techniques such as error weighted cross-correlation, PCA, RPCA with AdaBoost. Li et al.. (2018a; b) devised a FER model for partially occluded faces using attention based convolutional neural network mechanism (ACNN) which is not part-based or holistic-based. This model embedded two approaches pACNN and gACNN as an enhancement to the end-to-end trainable Gate Unit (GU) of CNN. pACNN introduced patch-based GU whereas gACNN introduced locality-

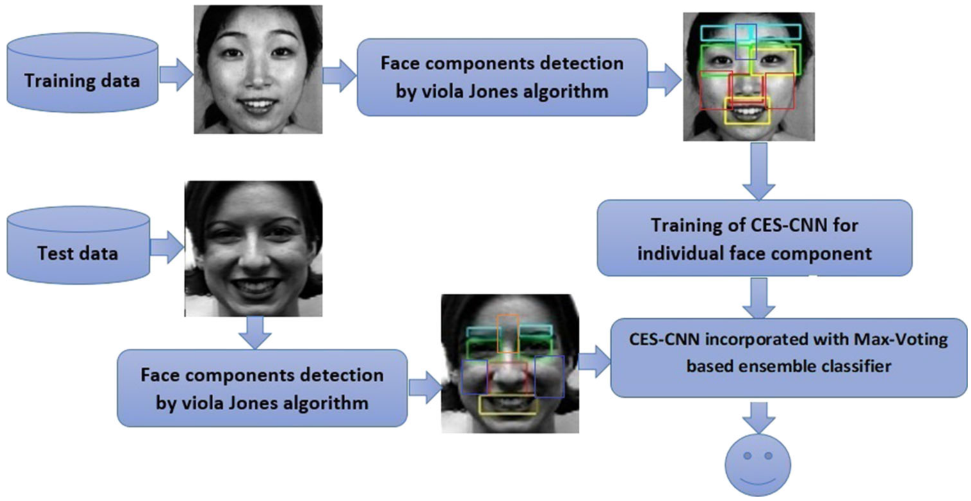


Fig. 3 Framework of the proposed model

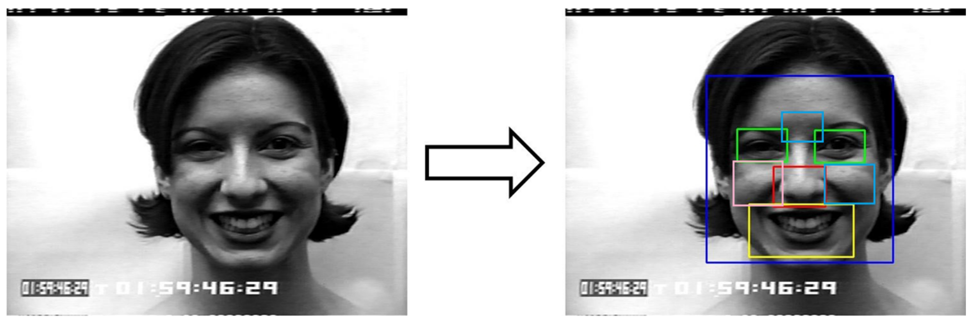


Fig. 4 Face components detected by Viola-Jones algorithm

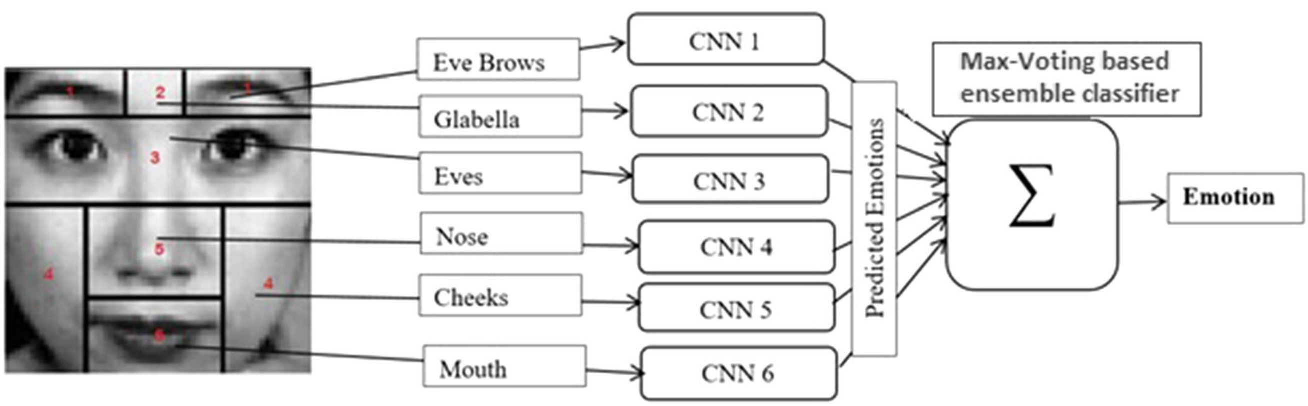


Fig. 5 The architecture of proposed CES-CNN model

based GU to assign weights based on unobstructedness in paths and global representations respectively.

Research gaps

The holistic-based FER models developed for recognizing emotions from partially occluded faces are more prone to reduced performance since they are not robust to the changed distortion conditions in the facial images. The part-based FER models which used generative approaches to replace the missing parts are biased to the occluded conditions. Furthermore, the part-based FER models neglect the missing parts needed to detect occluded faces exceptionally from every facial image. The solution to alleviate these issues is to develop a FER model which can recognize emotions more accurately with available facial parts. Hence, the component based Ensemble Stacked CNN is proposed for individual face components which can detect the facial emotion accurately even some parts of faces are missed due to occlusions by objects or illuminations or pose variations. The choice of ensemble stacked CNN is done in the proposed model to get the benefits such

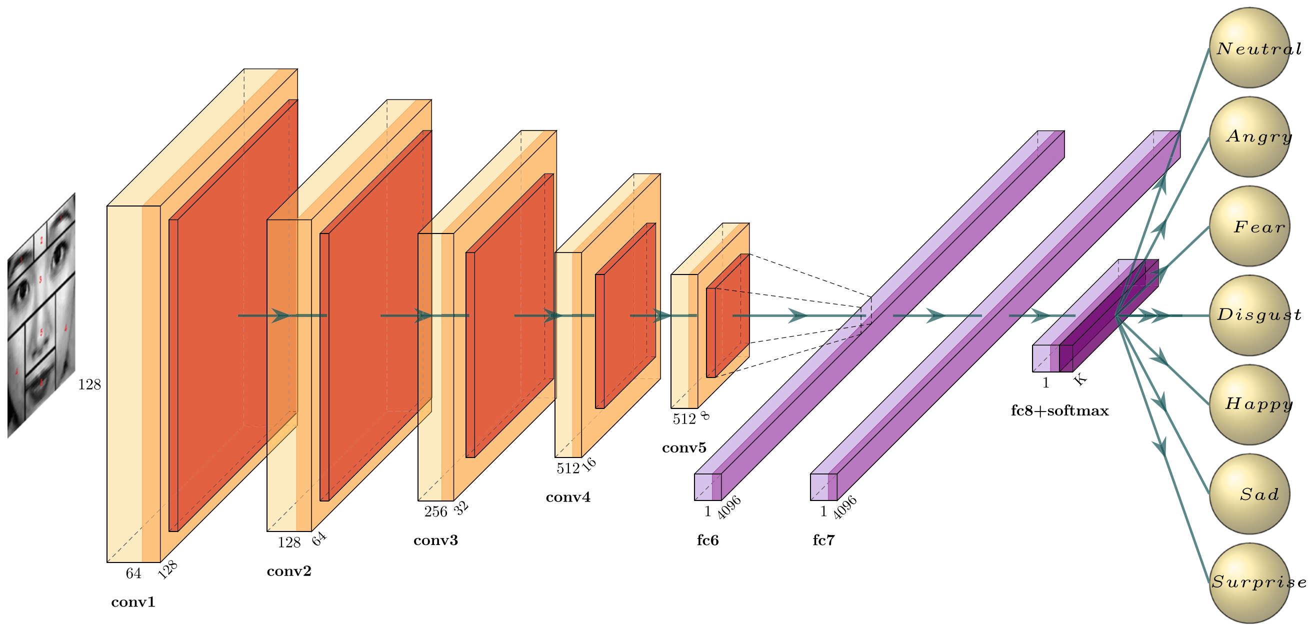


Fig. 6 Typical architecture of a CNN subnet

as preprocessing independent performance, in-built dimensionality reduction, automated feature extraction, enhanced performance, robust to huge data, ease to implementation, resistance to over-fitting problems. And particularly, it has been proved in many works that Ensemble CNN has superior performance than the remaining approaches.

Proposed model

Our proposed model aims to mitigate the issues of various distorted aspects that generally exist in the facial image taken in real conditions. The proposed model mainly consists of three stages: face components detection using Viola-Jones algorithm, applying CES-CNN for face components to analyze the emotions from each face component, and incorporating Max-Voting ensemble classifier in CES-CNN to increase the robustness in recognizing spontaneous facial expressions. The framework of the proposed model is as depicted in Fig. 3.

Face components detection using Viola-Jones algorithm

Viola-Jones algorithm is the most predominant algorithm developed for object detection and face components detection. In the proposed model, the Viola-Jones algorithm is used for detecting face components from partially occluded facial images. The Haar like features technique in the Viola-Jones algorithm is used in the approach to detect different face components as a rectangular box in the determined facial images. The face components detection algorithm considered for the proposed model can contend with different distortions like illumination, occlusions, and pose changes or rotated faces by using window classifiers of the Viola-Jones algorithm. Fig. 4 depicts different face components that can be detected by the Viola-Jones algorithm using Haar like features mechanism. The face components detected by the Viola-Jones algorithm are in different sizes. To maintain uniformity while training using CNN subnets, all the face components are rescaled to the same size.

Component based Ensemble Stacked CNN (CES-CNN)

Ensemble of classifiers is a set of classifiers whose individual decisions are combined in some way to classify new examples. Ensemble methods are more accurate than any individual members. A stack of convolutional neural networks (CNNs) is used in the proposed model to overcome the drawbacks found in traditional FER systems that are applied on a single modality (entire face) such as lighting conditions, pose changes, occlusions, and extracting features completely relying on the entire face image. The Component based Ensemble Stacked CNN (CES-CNN) proposed in the model considers individual facial parts such as eyebrows, eyes, nose, and mouth as features to train separate CNN classifiers where each classifier may predict different emotions for a particular facial component. The architecture of CES-CNN is as depicted in Fig. 5 and the

Table 1 Hyperparameters considered in proposed CES-CNN

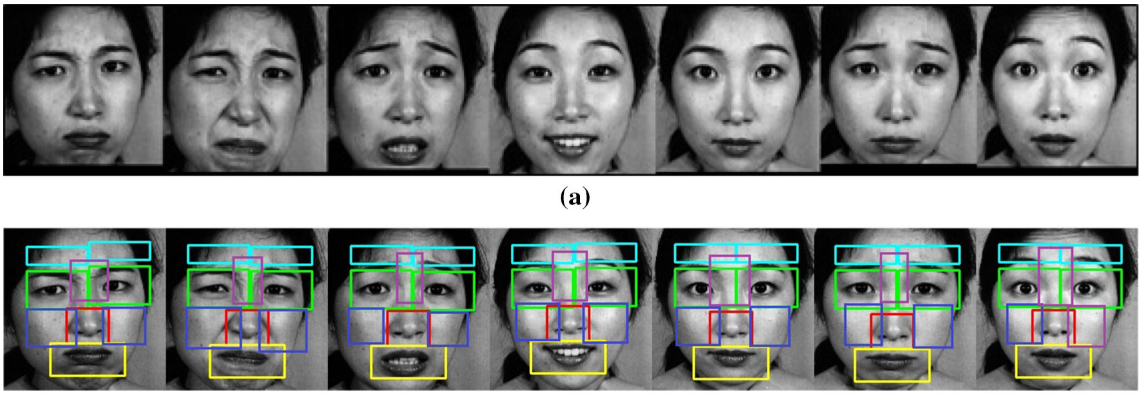
Block #	Layer	Output shape	Image size	Parameters
Block-1 (Conv-1)	Convolution2D (3 × 3)@64	(n,n,64)	(128,128,64)	((3 × 3)+1)×64 = 640
	Batch Normalization	(n,n,64)	(128,128,64)	4 × 64=256
	Activation Relu	(n,n,64)	(128,128,64)	0
	Maxpooling2D (2 × 2)	(n1,n1,64) n1=n/2	(64,64,64)	0
	Dropout	(n1,n1,64)	(64,64,64)	0
Block-2 (Conv-2)	Convolution2D (3 × 3)@128	(n1,n1,128)	(64,64,128)	((3 × 3×64)+1) ×128=73876
	Batch Normalization	(n1,n1,128)	(64,64,128)	4 × 128=512
	Activation Relu	(n1,n1,128)	(64,64,128)	0
	Maxpooling2D (2 × 2)	(n2,n2,128) n2=n1/2	(32,32,128)	0
	Dropout	(n2,n2,128)	(32,32,128)	0
Block-3 (Conv-3)	Convolution2D (3 × 3)@256	(n2,n2,256)	(32,32,256)	((3 × 3×128)+1) ×256=295168
	Batch Normalization	(n2,n2,256)	(32,32,256)	4 × 256=1024
	Activation Relu	(n2,n2,128)	(32,32,256)	0
	Maxpooling2D (2 × 2)	(n3,n3,256) n3=n2/2	(16,16,256)	0
	Dropout	(n3,n3,256)	(16,16,256)	0
Block-4 (Conv-4)	Convolution2D (3 × 3)@512	(n3,n3,512)	(16,16,512)	((3 × 3×256)+1) ×512=1180160
	Batch Normalization	(n3,n3,256)	(16,16,512)	4 × 512=2048
	Activation Relu	(n3,n3,256)	(16,16,512)	0
	Maxpooling2D (2 × 2)	(n4,n4,512) n4=n3/2	(8,8,512)	0
	Dropout	(n4,n4,256)	(8,8,512)	0
Block-5 (Conv-5)	Convolution2D (3 × 3)@512	(n4,n4,512)	(8,8,512)	((3 × 3×512)+1) ×512=2359808
	Batch Normalization	(n4,n4,512)	(8,8,512)	4 × 512=2048
	Activation Relu	(n4,n4,512)	(8,8,512)	0
	Maxpooling2D (2 × 2)	(n5,n5,512) n5=n4/2	(4,4,512)	0
	Dropout	(n5,n5,512)	(4,4,512)	0
	Flatten	(1,$n5$ × $n5$×512	(1,8192)	0
	Dense	(1256)	(1,256)	(8192+1)×256 = 2097408
	Batch Normalization	(1,256)	(1,256)	512
	activation Relu	(1,256)	(1,256)	0
	Dropout	(1,256)	(1,256)	0
	Dense	(1,512)	(1,512)	(256+1)×512 = 131584
	Batch Normalization	(1,512)	(1,512)	1024
	activation Relu	(1,512)	(1,512)	0
	Dropout	(1,512)	(1,512)	0
	Dense	(1,7)	(1,7)	(512+1)×7 = 3591
	Total parameters for image of size (128 × 128)			9481607

typical architecture of each CNN subnet is as depicted in Fig. 6.

In the proposed CES-CNN, VGG16 based convolutional neural network (CNN) architecture is used to develop each subnet. The CNN used for developing the subnets contain the core building blocks of convolutional layer (Vedaldi and Zisserman 2016), max-pooling layer (Vedaldi and Zisserman 2016), fully connected layer (Targ et al. 2016) and dense layer (Targ et al. 2016). Additionally, for improving the stability, performance, and speed of the CNN, the batch normalization layer has also been included in the proposed CNN architecture.

Hyper parameters of the CNN subnet

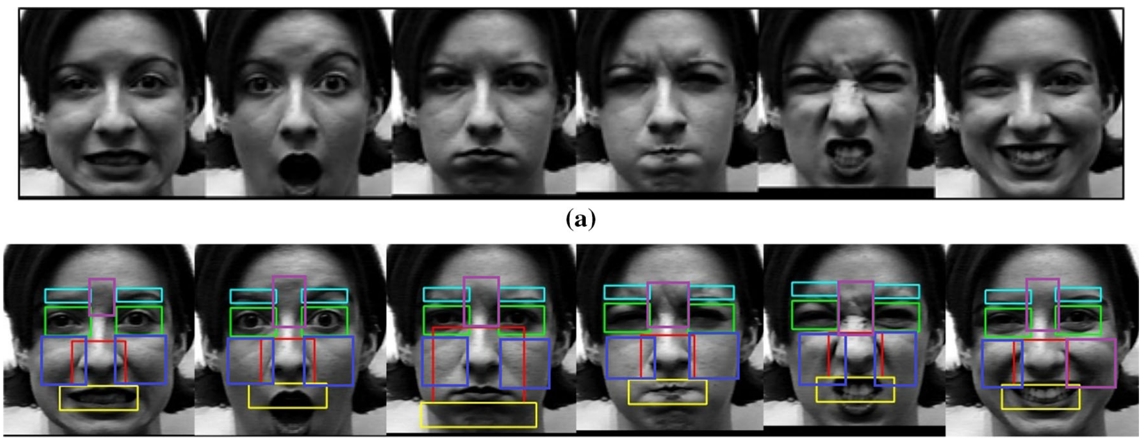
The hyper parameters described in Table 1 are used in each of the CNN subnets of the CES-CNN model along with the other parameters such as learning rate, batch size, epochs



(a)

(b)

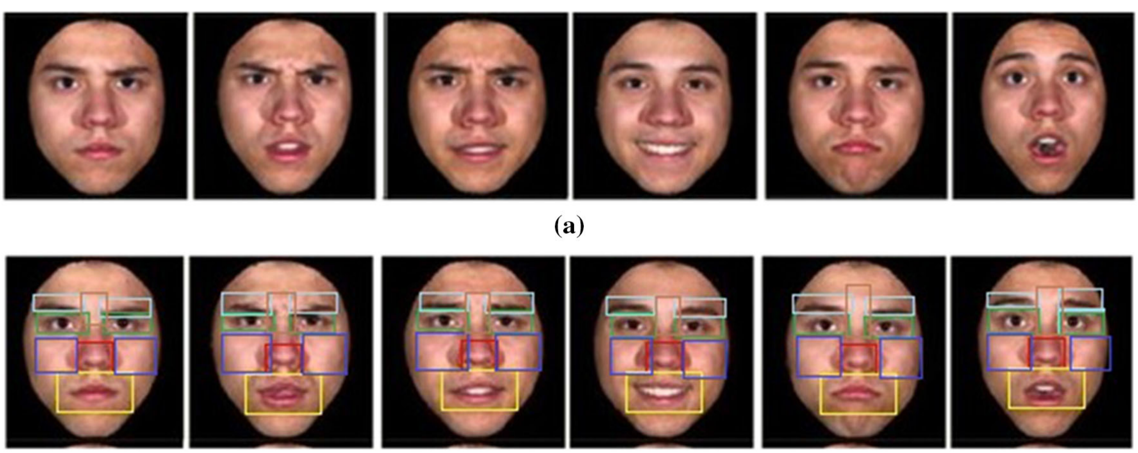
Fig. 7 Sample face images **a** resized **b** face components detected Viola-Jones from JEFFE dataset



(a)

(b)

Fig. 8 Sample face images **a** resized **b** face components detected Viola-Jones from CK+ dataset



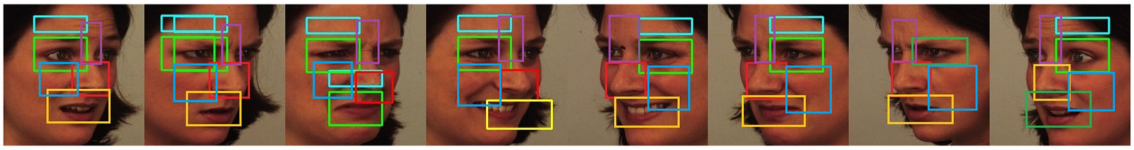
(a)

(b)

Fig. 9 Sample face images **a** resized **b** face components detected Viola-Jones from BU-3DFE dataset

which are extensively discussed in the results and discussion section. The face components detected by the Viola-Jones algorithm are in different sizes. To maintain uniformity while training using CNN subnets, all the face components are rescaled to the same size.

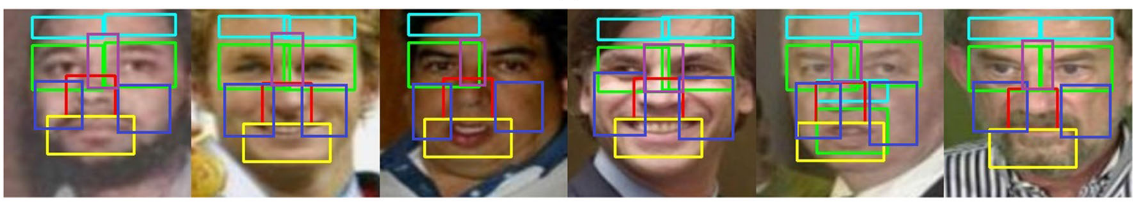
(a)



(b)

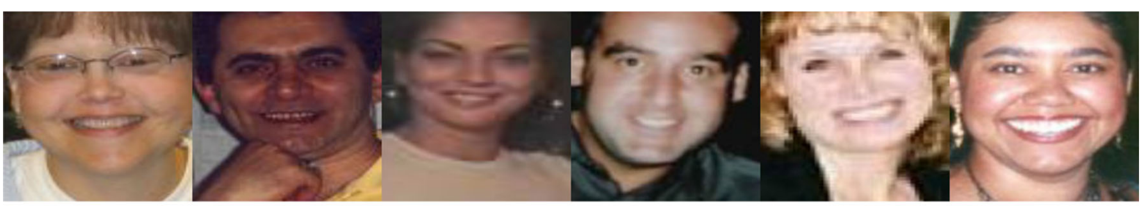
Fig. 10 Sample face images **a** resized **b** face components detected Viola-Jones from KDEF dataset

(a)

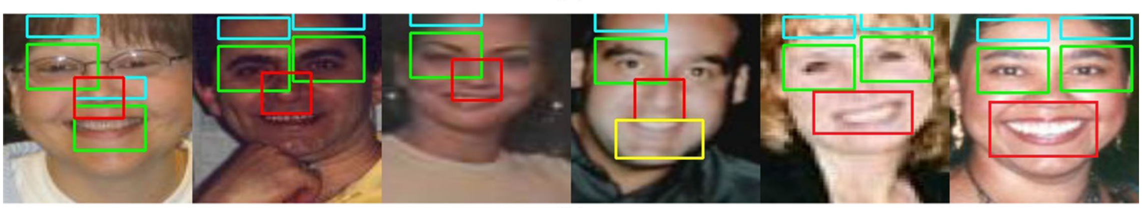


(b)

Fig. 11 Sample face images **a** resized **b** face components detected Viola-Jones from LFW dataset



(a)



(b)

Fig. 12 Sample face images **a** resized **b** face components detected Viola-Jones from GENKI4K dataset

Max-Voting classifier

Max-Voting (majority Voting) is an ensemble machine learning algorithm. It is proven that a majority voting ensemble improved the model performance than any of the single models used in the ensemble. There are two types of voting algorithms: Soft Voting and Hard Voting. Soft voting predicts the class with the highest sum of probabilities, but the individual CES-CNN models output a class label here, so it cannot be used. A hard voting ensemble mechanism is used here, which takes the sum of the votes

Table 2 Evaluation metrics on various datasets using CES-CNN

Parameter	Dataset					
	JAFFE	CK+	KDEF	BU-3DFE	LFW	GENKI
Accuracy	95.39	96.64	94.42	96.24	91.33	98.56
Precision	95.44	96.67	94.48	96.27	91.48	98.48
Recall	95.39	96.64	94.42	96.24	91.33	98.56
F1-Score	95.40	96.65	94.43	96.25	91.36	98.36

from the individual CES-CNNs for each of the components and outputs the class label with maximum votes.

Results and analysis

Five openly accessible benchmark datasets viz JAFFE, Cohn Kanade, KDEF, 3D-BUFE, and LFW emotion databases are utilized for experimenting and evaluating the proposed model. The actual pictures from these datasets are resized to 128x128 pixels to maintain uniformity on the training algorithm.

The Japanese Female Facial Expression (JAFFE) database (Lyons et al. 1998) has a total of 213 still images posed by 10 Japanese models and has 7 different expressions including neutral. The dataset is relatively small and was used in the prototyping stages. The sample images that are resized are shown in Fig. 7a, whereas Fig. 7b depicts the components detected by the Viola-Jones in each facial image of JAFFE.

The Cohn-Kanade dataset (Kanade et al. 2000) is one of the standard datasets for research in automatic facial expression classification and emotion prediction. There are two versions available for the Cohn-Kanade dataset, CK and CK+. There are a total of 593 facial images of 123 participants in the dataset and each image is 640x490 in size. The faces in CK and CK+ datasets are captured in low illumination conditions. The sample face images that are resized to 128x128 and the face components detected by the Viola-Jones algorithm are as shown in Fig. 8.

Binghamton University 3D Facial Expression (BU-3DFE) (Yin et al. 2006) contains 2500 color images covering 7 basic emotions of 100 subjects aged ranging from 18 to 70 and interracial ethnicity. It also contains corresponding facial texture images captured at two views (+45° and −45°) and geometric shape models. The face images of BU-3DFE that are resized to 128x128 and the face components detected by the Viola-Jones algorithm on each face image are shown in Fig 9.

Karolinska Directed Emotional Faces (KDEF) (Calvo and Lundqvist 2008) dataset consists of 4900 color images of size 562x762 posing seven basic expressions from 70 subjects. Four hundred images of the seven basic expression types were considered for experimentation where faces images are resized to 128x128 and face components are detected using the Viola-Jones are as shown in Fig 10.

Labeled Faces in the Wild (LFW) (Huang et al. 2008) is intended for face recognition tasks, however, due to its diversity, low resolution, partially occluded nature, poor lighting, and wide range of age differences of subjects, it is a challenging dataset for the facial emotion recognition task. It contains 13000 color images collected from the web. The sample face images of LFW that are resized to 128x128 and face components detected using the Viola-Jones are shown in Fig 11.

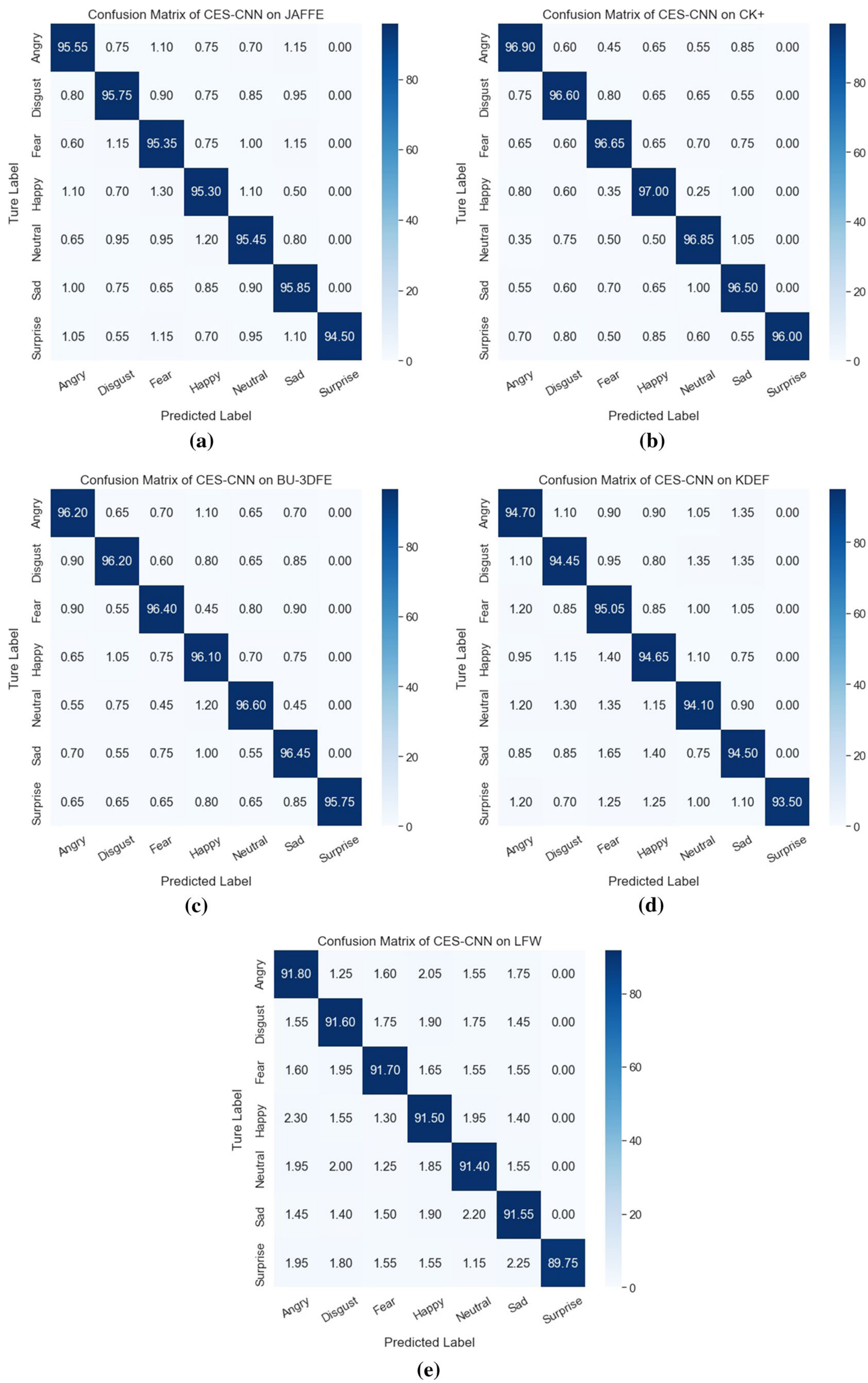
GENKI (Jain and Crowley 2013) is a Face, Expression, and Pose Dataset consisting of 4000 images of a wide range of illumination conditions, geographical locations, personal identity, and ethnicity. It contains two class labels for expressions of smiling and non-smiling faces. The dataset consists of images of size 179x192. It is a challenging dataset for expression recognition considering its characterictics. The sample face images and images of face components detected using the Viola-Jones are shown in Fig 12.

Performance of the proposed model on individual datasets

Many performance measures are used by researchers such as classification accuracy, precision, recall, F-measure (Powers 2020). In performance evaluation on individual datasets, the training data is constructed on the same dataset as the testing dataset in this test. The data has been shuffled before splitting it into training and testing sets. Tenfold cross validation was employed on these tests to rigorously test the proposed models. The seven expressions Angry (AN), Sad(SA), Fear (FE), Disgust (DI), Happy (HA), Neutral (NE) and Surprise (SU) are recognized from test images using proposed CES-CNN as they are the most common expressions recognized by FER systems. The classification accuracies of the proposed model are recorded on individual datasets and presented in Table 2.

The performance results of the proposed model on an individual dataset with regards to the confusion matrices are given in Fig. 13.

And finally, the proposed model is also tested for generalization by preparing the training data from all five datasets. The dataset is shuffled and split into training and testing sets and the average recognition rate is calculated as the accuracy in recognizing the facial expression of a face considered from a combined dataset. The average recognition rate results are produced in Table 3 and it can be clearly observed from the results that the accuracies are in


Confusion Matrix of CES-CNN on JAFFE
Confusion Matrix of CES-CNN on CK+
Confusion Matrix of CES-CNN on BU-3DFE
Confusion Matrix of CES-CNN on KDEF
Confusion Matrix of CES-CNN on LFW
Ture Label
Predicted Label
Angry
Disgust
Fear
Happy
Neutral
Sad
Surprise
(a)
(b)
(c)
(d)
(e)

◀**Fig. 13** Performance evaluation of CES-CNN using confusion matrix on **a** JAFFE dataset **b** CK+ dataset **c** BU-3DFE dataset **d** KDEF dataset **e** LFW dataset

Table 3 Average recognition rates of the proposed CES-CNN model on combined dataset

Dataset	Average recognition rate (%)
JAFFE	95.39
CK+	96.64
KDEF	94.42
BU-3DFE	96.24
LFW	91.33
GENKI	98.56

the decreasing order on the datasets in the order GENKI, CK+, BU-3DFE, JAFFE, KDEF, and LFW respectively. It is also evident from the precision and recall values that these models are more generalized and are proved from the values of accuracy and F1-score.

Performance of the proposed model for various hyper parameters

The proposed model is also trained and rigorously tested by varying the hyper parameters. The model is trained at different learning rates ranging from 0.01, 0.001, 0.0001, and 0.00001. The accuracies obtained for these settings are depicted in Fig. 14. The learning rate has impacted the model training on certain datasets and hence affected the recognition accuracy as shown in Fig. 14.

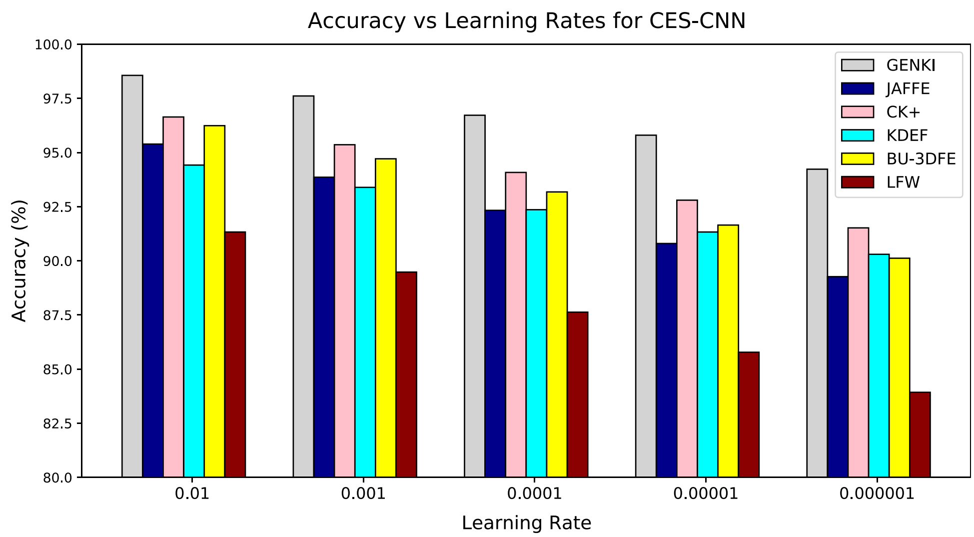


Fig. 14 Comparison of validation accuracy for different learning rates

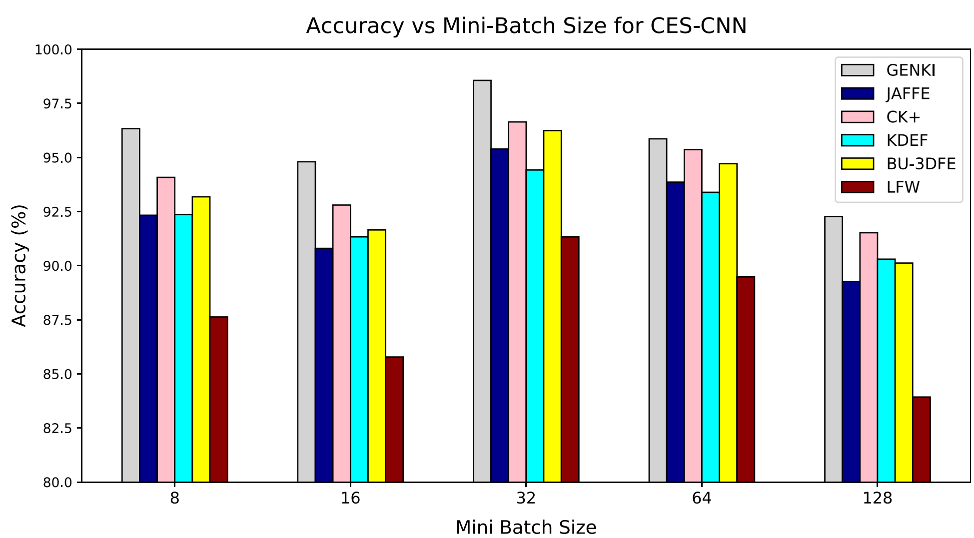


Fig. 15 Comparison of validation accuracy for different mini batch sizes

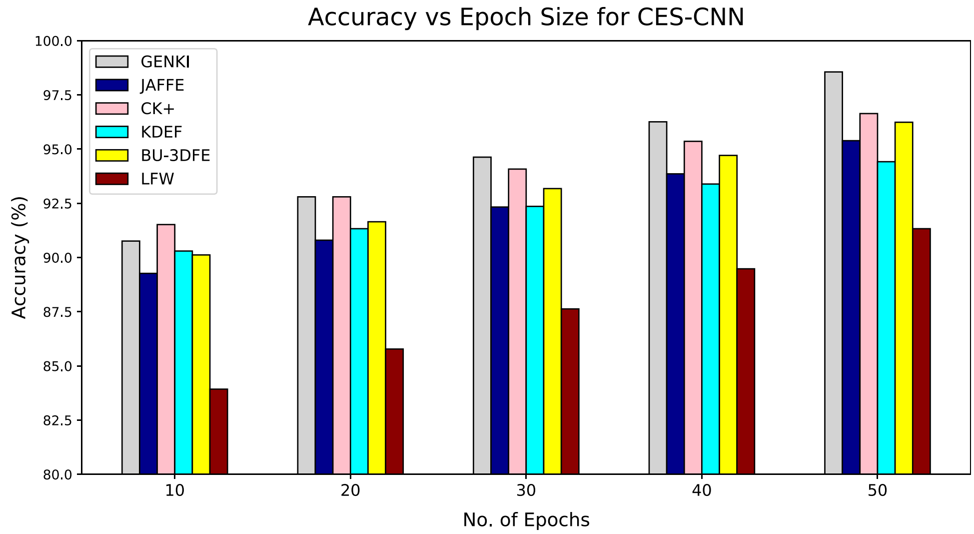


Fig. 16 Comparison of validation accuracy for different epochs

Fig. 17 Synthetic face part occlusion images from JAFFE-O and CK+-O dataset

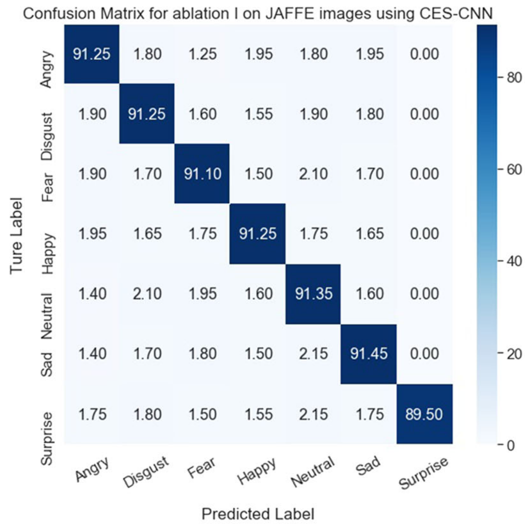
Fig. 18 Facial images **a** depict the scenario of ablation I **b** masked over the mouth component

The recognition accuracies of the proposed model are recorded for varying batch sizes as shown in Fig. 15. Though the validation accuracy has fluctuated over the mini-batch sizes for some datasets, GENKI, JAFFE, CK+, and KDEF datasets maintained stability on the same settings. It can be observed from Fig. 15 that the model achieved the best generalization for a batch size of 32.

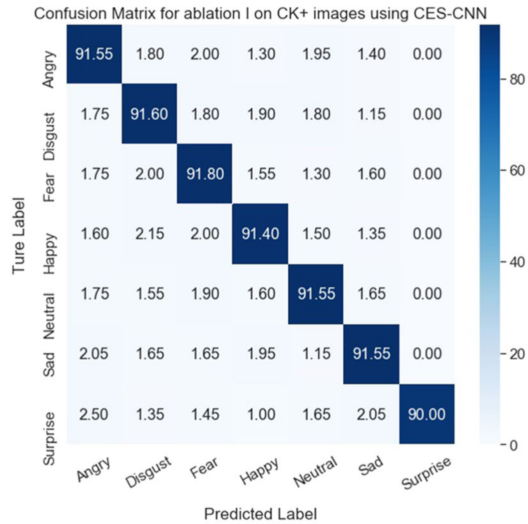
Apart from the learning rate and mini-batch size, the number of training epochs is also an important criterion. Though stacked CNN training takes longer than usual, these models were trained for a maximum of 50 epochs and the validation accuracies are recorded for epochs at 10, 20, 30, 40, and 50. Fig. 16 shows the bar chart of the recorded values and it is clear that the model achieved higher accuracies for 50 epochs than the remaining.

Performance of the proposed model on partially occluded faces

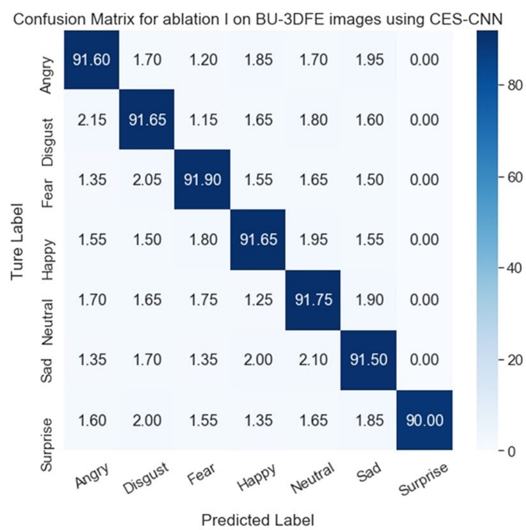
Facial emotion recognition in the wild is a challenging task. Many forms of occlusion cause the noise in face data preventing the FER system from identifying a face and subsequently expression. Although LFW and KDEF dataset contains face images of limited occlusion, i.e rotated to one side. However, the datasets JAFFE, CK+, and BU-3DFE facial images are taken under laboratory conditions and thus have no occlusions. Hence, synthetic occlusions are generated for GENKI, JAFFE, CK+, BU-3DFE and are


Confusion Matrix for ablation I on JAFFE images using CES-CNN
Ture Label
Angry 91.25 1.80 1.25 1.95 1.80 1.95 0.00
Disgust 1.90 91.25 1.60 1.55 1.90 1.80 0.00
Fear 1.90 1.70 91.10 1.50 2.10 1.70 0.00
Happy 1.95 1.65 1.75 91.25 1.75 1.65 0.00
Neutral 1.40 2.10 1.95 1.60 91.35 1.60 0.00
Sad 1.40 1.70 1.80 1.50 2.15 91.45 0.00
Surprise 1.75 1.80 1.50 1.55 2.15 1.75 89.50
Angry Disgust Fear Happy Neutral Sad Surprise
Predicted Label
80 60 40 20 0

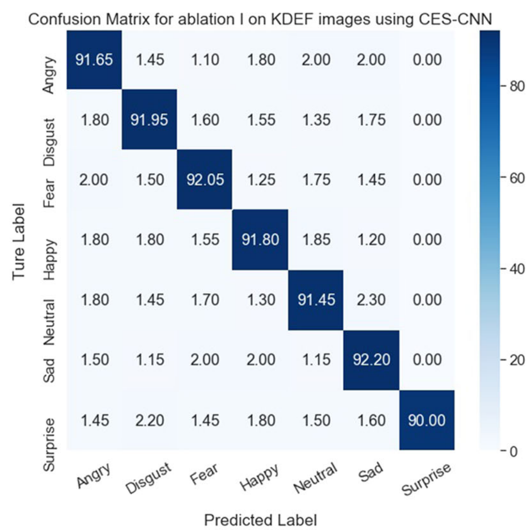
(a)


Confusion Matrix for ablation I on CK+ images using CES-CNN
Ture Label
Angry 91.55 1.80 2.00 1.30 1.95 1.40 0.00
Disgust 1.75 91.60 1.80 1.90 1.80 1.15 0.00
Fear 1.75 2.00 91.80 1.55 1.30 1.60 0.00
Happy 1.60 2.15 2.00 91.40 1.50 1.35 0.00
Neutral 1.75 1.55 1.90 1.60 91.55 1.65 0.00
Sad 2.05 1.65 1.65 1.95 1.15 91.55 0.00
Surprise 2.50 1.35 1.45 1.00 1.65 2.05 90.00
Angry Disgust Fear Happy Neutral Sad Surprise
Predicted Label
80 60 40 20 0

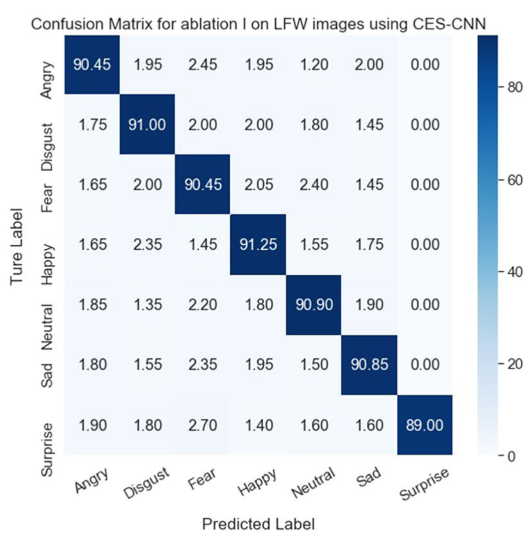
(b)


Confusion Matrix for ablation I on BU-3DFE images using CES-CNN
Ture Label
Angry 91.60 1.70 1.20 1.85 1.70 1.95 0.00
Disgust 2.15 91.65 1.15 1.65 1.80 1.60 0.00
Fear 1.35 2.05 91.90 1.55 1.65 1.50 0.00
Happy 1.55 1.50 1.80 91.65 1.95 1.55 0.00
Neutral 1.70 1.65 1.75 1.25 91.75 1.90 0.00
Sad 1.35 1.70 1.35 2.00 2.10 91.50 0.00
Surprise 1.60 2.00 1.55 1.35 1.65 1.85 90.00
Angry Disgust Fear Happy Neutral Sad Surprise
Predicted Label
80 60 40 20 0

(c)


Confusion Matrix for ablation I on KDEF images using CES-CNN
Ture Label
Angry 91.65 1.45 1.10 1.80 2.00 2.00 0.00
Disgust 1.80 91.95 1.60 1.55 1.35 1.75 0.00
Fear 2.00 1.50 92.05 1.25 1.75 1.45 0.00
Happy 1.80 1.80 1.55 91.80 1.85 1.20 0.00
Neutral 1.80 1.45 1.70 1.30 91.45 2.30 0.00
Sad 1.50 1.15 2.00 2.00 1.15 92.20 0.00
Surprise 1.45 2.20 1.45 1.80 1.50 1.60 90.00
Angry Disgust Fear Happy Neutral Sad Surprise
Predicted Label
80 60 40 20 0

(d)


Confusion Matrix for ablation I on LFW images using CES-CNN
Ture Label
Angry 90.45 1.95 2.45 1.95 1.20 2.00 0.00
Disgust 1.75 91.00 2.00 2.00 1.80 1.45 0.00
Fear 1.65 2.00 90.45 2.05 2.40 1.45 0.00
Happy 1.65 2.35 1.45 91.25 1.55 1.75 0.00
Neutral 1.85 1.35 2.20 1.80 90.90 1.90 0.00
Sad 1.80 1.55 2.35 1.95 1.50 90.85 0.00
Surprise 1.90 1.80 2.70 1.40 1.60 1.60 89.00
Angry Disgust Fear Happy Neutral Sad Surprise
Predicted Label
80 60 40 20 0

(e)

◀ **Fig. 19** Performance evaluation of CES-CNN using confusion matrix for ablation I on **a** JAFFE-O dataset **b** CK+-O dataset **c** BU-3DFE-O **d** KDEF-O dataset **e** LFW-O dataset

Fig. 20 Facial images **a** depict the scenario of ablation II **b** masked over the mouth and eyes component

named GENKI-O, JAFFE-O, CK+-O, BU-3DFE-O respectively. The synthetic occlusions generated for JAFFE and CK+ are shown in Fig. 17. These synthetic occlusions are generated by finding face components using the Viola-Jones algorithm and then randomly masking face components with boxes of white pixels.

A minimum of one component and a maximum of three components in a single image are blocked in the automatic generation of the synthetic dataset. The partially covered face regions will not provide sufficient information to train the CNN for recognizing the facial expression from the corresponding face region. Hence, such a scenario is not considered for the evaluation of the proposed model. The number of occluded parts is considered random, i.e, a fixed number of occluded parts are not considered every time for experimentation. The number of occluded parts is considered in the range of 1 to the maximum of 3 parts out of the six parts. If it is the case, the face is fully rotated to one side, the Viola-Jones algorithm detected three components such as eye, eyebrows, and cheek.

It is challenging for any FER system to recognize emotions from an occluded face such as shown in Fig. 17. A traditional CNN model would fail to detect any emotion from such an occluded image. Specialized models are required to address these problems hence CES-CNN is proposed. Rigorous experiments are conducted on these occluded images and values for various evaluation metrics are recorded.

Ablation Study

An ablation study is conducted to exploit the robustness of the proposed model, where facial components such as eyes, nose, mouth, cheeks, eyebrows, or glabella are occluded. When a particular face component cannot be detected due to occlusions such as a person wearing a hat, mouth covered with arms when laughing, eyes covered with protective glasses or person turns over showing only half face, any of the occluded face components cannot be extracted and hence the respective CNN subnet in the proposed model will not produce any emotion recognition output. Then the proposed model accuracy is the ensemble results of emotion classification obtained from the remaining CNN subnets which are provided with the non-occluded face components. The ablation study is conducted for three scenarios. In the first scenario, any one of the facial components is considered to have occlusion. In the second scenario, any two of the facial components are considered occluded. In the third scenario, any three of the facial components are considered to have occlusion.

Ablation I: Occlusion on single face component The case study considered as the ablation I is one face component is occluded in such a scenario that the person covered the mouth with arms while giving expression. Sample image is shown in Fig. 18a which depicts this scenario and the facial image of the dataset masked over the mouth component is shown in Fig. 18b.

Experiments are performed on each dataset considering this scenario and model recognition performances are depicted in the confusion matrices shown in Fig. 19a through Fig. 19e.

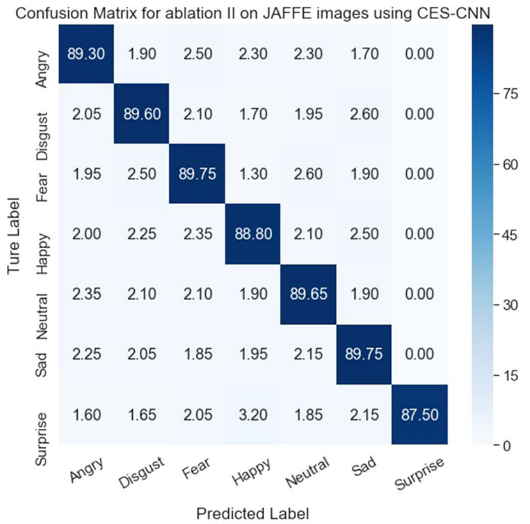
Ablation II: Occlusion on two face component The case study considered as the ablation II is two face components are occluded simultaneously in such a scenario that a person wearing sunglasses covers the mouth while giving expression. Sample image is shown in Fig. 20a which depicts this scenario and the facial image of the dataset masked over the mouth component and eyes component is shown in Fig. 20b.

Experiments are performed on each dataset considering this scenario and model recognition performances are depicted in the confusion matrices shown in Fig. 21a through Fig. 21e.

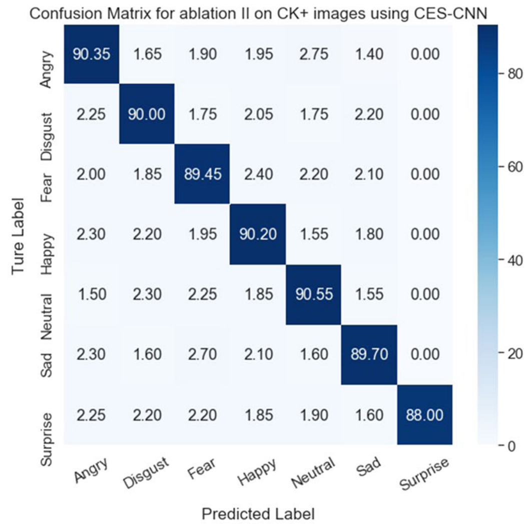
Ablation III: Occlusion on three face component The case study considered as the ablation III is three face components are occluded simultaneously in such a scenario that a person wearing a mask covering mouth, nose, and cheeks components. Sample image is shown in Fig. 22a which depicts this scenario and the facial image of the dataset masked over the mouth, nose, and cheeks components are shown in Fig. 22b.

Experiments are performed on each dataset considering this scenario and model recognition performances are depicted in the confusion matrices shown in Fig. 23a through Fig. 23e.

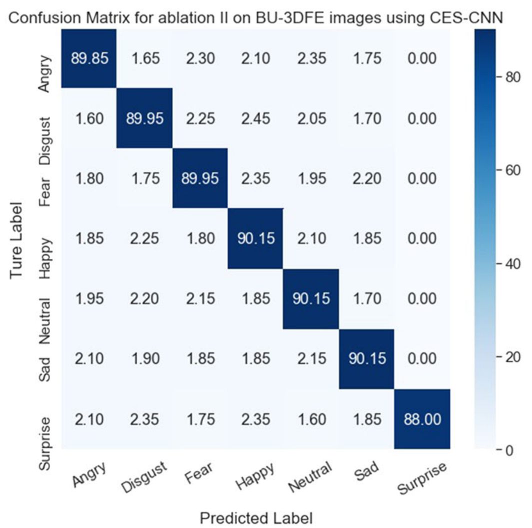
It is observed from the ablation study that the performance of the model is decreasing with the increased


Confusion Matrix for ablation II on JAFFE images using CES-CNN
Ture Label
Predicted Label
Angry
Disgust
Fear
Happy
Neutral
Sad
Surprise
89.30 1.90 2.50 2.30 2.30 1.70 0.00
2.05 89.60 2.10 1.70 1.95 2.60 0.00
1.95 2.50 89.75 1.30 2.60 1.90 0.00
2.00 2.25 2.35 88.80 2.10 2.50 0.00
2.35 2.10 2.10 1.90 89.65 1.90 0.00
2.25 2.05 1.85 1.95 2.15 89.75 0.00
1.60 1.65 2.05 3.20 1.85 2.15 87.50
75
60
45
30
15
0

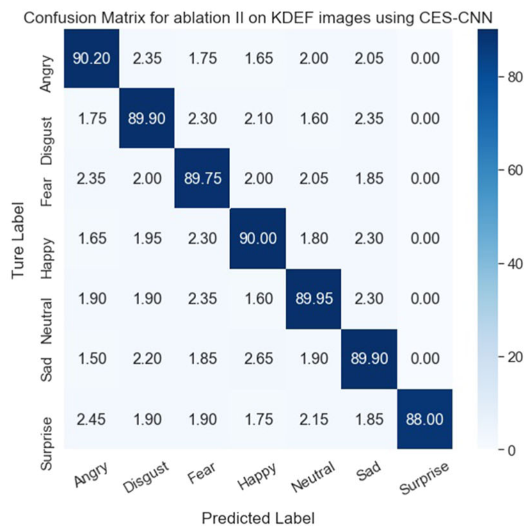
(a)


Confusion Matrix for ablation II on CK+ images using CES-CNN
Ture Label
Predicted Label
Angry
Disgust
Fear
Happy
Neutral
Sad
Surprise
90.35 1.65 1.90 1.95 2.75 1.40 0.00
2.25 90.00 1.75 2.05 1.75 2.20 0.00
2.00 1.85 89.45 2.40 2.20 2.10 0.00
2.30 2.20 1.95 90.20 1.55 1.80 0.00
1.50 2.30 2.25 1.85 90.55 1.55 0.00
2.30 1.60 2.70 2.10 1.60 89.70 0.00
2.25 2.20 2.20 1.85 1.90 1.60 88.00
80
60
40
20
0

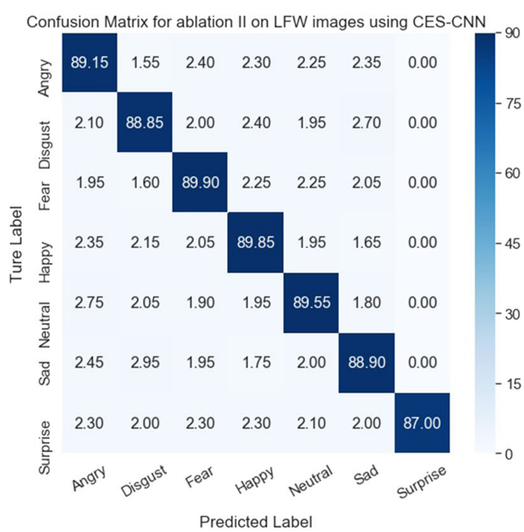
(b)


Confusion Matrix for ablation II on BU-3DFE images using CES-CNN
Ture Label
Predicted Label
Angry
Disgust
Fear
Happy
Neutral
Sad
Surprise
89.85 1.65 2.30 2.10 2.35 1.75 0.00
1.60 89.95 2.25 2.45 2.05 1.70 0.00
1.80 1.75 89.95 2.35 1.95 2.20 0.00
1.85 2.25 1.80 90.15 2.10 1.85 0.00
1.95 2.20 2.15 1.85 90.15 1.70 0.00
2.10 1.90 1.85 1.85 2.15 90.15 0.00
2.10 2.35 1.75 2.35 1.60 1.85 88.00
80
60
40
20
0

(c)


Confusion Matrix for ablation II on KDEF images using CES-CNN
Ture Label
Predicted Label
Angry
Disgust
Fear
Happy
Neutral
Sad
Surprise
90.20 2.35 1.75 1.65 2.00 2.05 0.00
1.75 89.90 2.30 2.10 1.60 2.35 0.00
2.35 2.00 89.75 2.00 2.05 1.85 0.00
1.65 1.95 2.30 90.00 1.80 2.30 0.00
1.90 1.90 2.35 1.60 89.95 2.30 0.00
1.50 2.20 1.85 2.65 1.90 89.90 0.00
2.45 1.90 1.90 1.75 2.15 1.85 88.00
80
60
40
20
0

(d)


Confusion Matrix for ablation II on LFW images using CES-CNN
Ture Label
Predicted Label
Angry
Disgust
Fear
Happy
Neutral
Sad
Surprise
89.15 1.55 2.40 2.30 2.25 2.35 0.00
2.10 88.85 2.00 2.40 1.95 2.70 0.00
1.95 1.60 89.90 2.25 2.25 2.05 0.00
2.35 2.15 2.05 89.85 1.95 1.65 0.00
2.75 2.05 1.90 1.95 89.55 1.80 0.00
2.45 2.95 1.95 1.75 2.00 88.90 0.00
2.30 2.00 2.30 2.30 2.10 2.00 87.00
90
75
60
45
30
15
0

(e)

◀ **Fig. 21** Performance evaluation of CES-CNN using confusion matrix for ablation II on **a** JAFFE-O dataset **b** CK+-O dataset **c** BU-3DFE-O **d** KDEF-O dataset **e** LFW-O dataset

Fig. 22 Facial images **a** depict the scenario of ablation III **b** masked over the component on on side of the face

number of occluded facial parts as the facial expression recognition can only be performed with the CNN subsets in which corresponding components are not occluded. . To evaluate the generalization of the CES-CNN model, cross-database validation is performed on partially occluded faces in which two face components are masked. In our experimentation, the cross-database validation is conducted by considering the partially occluded facial images from any four datasets as training data and the remaining dataset as the test data. Table 4 shows values of cross-database validation in terms of various evaluation metrics such as accuracy, precision, recall, and F1-score for different test datasets. It is evident from the results that the proposed model achieved the best generalization for all partially occluded face images from datasets like JAFFE-O, CK+-O, KDEF-O, BU-3DFE-O, GENKI-O and LFW-O.

Comparison with the state-of-the-art

The proposed model CES-CNN is encompassing two features. Firstly, it is an ensemble model for enhancing the overall performance of FER. Second, it handles partial occlusions effectively. To analyze these two capabilities of the CES-CNN model, the comparisons were made in two categories.

Performance comparison with ensemble models

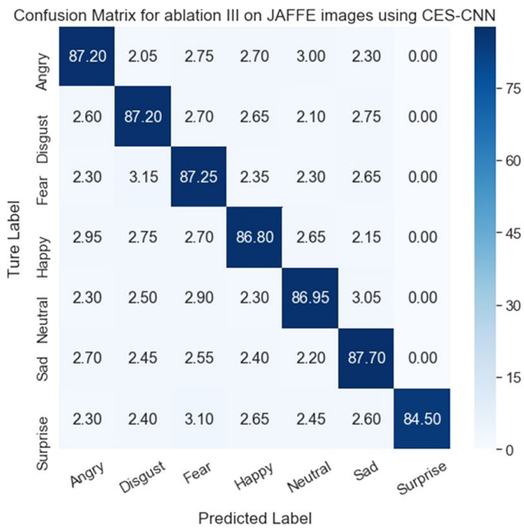
In this section, the average accuracies are compared between existing ensemble models and the proposed CES-CNN ensemble model. The accuracies of the proposed model on individual datasets are compared with the existing state-of-the-art models that are discussed in the literature. Tables 5, 6, 7, 8, and 9 shows the comparison results of the proposed model with the existing models on JAFFE, CK+, KDEF, BU-3DFE, and LFW datasets respectively. In this paper, images from all datasets were used in the training phase for better generalization of the model. In the experimental analysis of the proposed model, we merged all the images from different datasets considered for the experimentation for training phase. Our motive is to develop an FER model in such a way that it should be able to analyze any kind of facial image (taken within different illumination conditions or having different occlusion scenarios). We rigorously tested our model by considering test images individually from each dataset and calculated the performance of the model. The experimental results have shown the better performance on the images of all the datasets considered for the experimentation. It is evident from Tables 5 through 9 that the proposed CES-CNN model outperforms the existing models.

Performance comparison on partially occluded faces

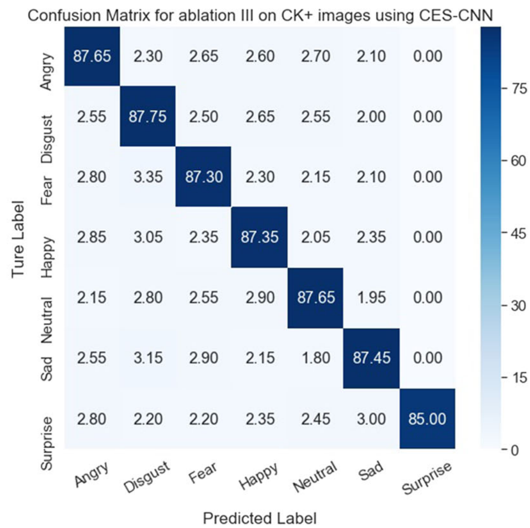
The proposed model CES-CNN is compared with the existing state-of-the-art models developed on partially occluded facial images. For synthetic datasets generated from JAFFE, CK+, and BU-3DFE, the performance of the proposed model and the existing models is measured by applying them on partially occluded faces in which two facial components are masked. Table 11 presents the accuracies on non-occluded and on occluded faces on JAFFE. Though the mean accuracy for non-occluded faces obtained in Liu et al. (2018) and Zia et al. (2018) is more than the proposed model, it is observed that the accuracy on occluded faces for the proposed model is high on the JAFFE dataset.

Table 12 presents the mean accuracies of the existing methods and the proposed model on non-occluded versus occluded face images on the synthesized CK+ dataset.

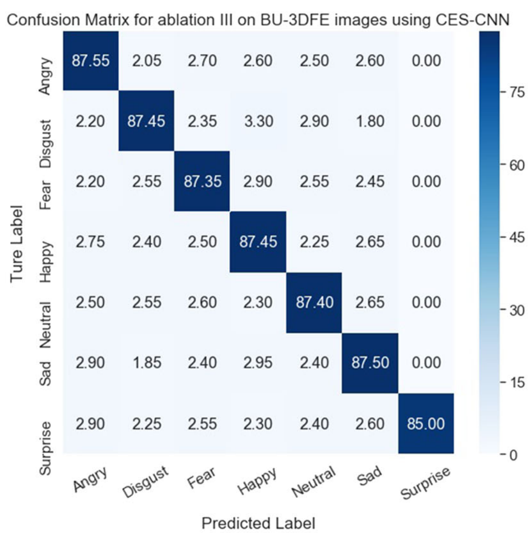
We can observe from these results of Table 12 that the average accuracies obtained in (Liu et al. 2018) and (Li et al. 2018a) records higher values than the proposed CES-CNN model on non-occluded face images. But, on the occluded face images, the proposed model outperforms the existing models. We can observe the comparison of the accuracies between existing and the proposed CES-CNN model on occluded versus non-occluded face images on the KDEF, BU-3DFE, and LFW datasets in Tables 13, 14, and 15 respectively. It is clear from the results presented in tables that the proposed CES-CNN model for FER on partially occluded faces outperforms the existing models.


Confusion Matrix for ablation III on JAFFE images using CES-CNN
Ture Label
Predicted Label
Angry Disgust Fear Happy Neutral Sad Surprise
87.20 2.05 2.75 2.70 3.00 2.30 0.00
2.60 87.20 2.70 2.65 2.10 2.75 0.00
2.30 3.15 87.25 2.35 2.30 2.65 0.00
2.95 2.75 2.70 86.80 2.65 2.15 0.00
2.30 2.50 2.90 2.30 86.95 3.05 0.00
2.70 2.45 2.55 2.40 2.20 87.70 0.00
2.30 2.40 3.10 2.65 2.45 2.60 84.50
0 15 30 45 60 75

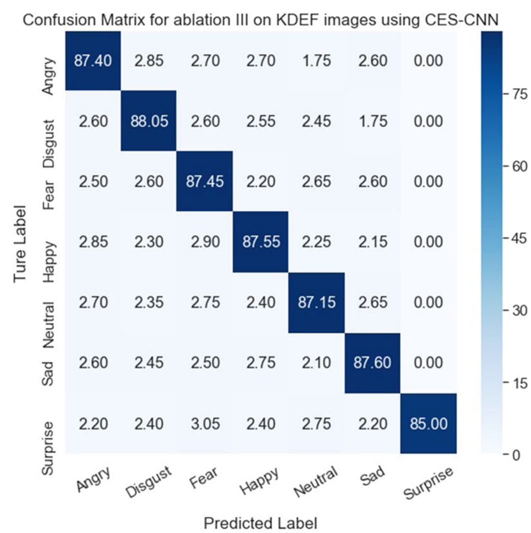
(a)


Confusion Matrix for ablation III on CK+ images using CES-CNN
Ture Label
Predicted Label
Angry Disgust Fear Happy Neutral Sad Surprise
87.65 2.30 2.65 2.60 2.70 2.10 0.00
2.55 87.75 2.50 2.65 2.55 2.00 0.00
2.80 3.35 87.30 2.30 2.15 2.10 0.00
2.85 3.05 2.35 87.35 2.05 2.35 0.00
2.15 2.80 2.55 2.90 87.65 1.95 0.00
2.55 3.15 2.90 2.15 1.80 87.45 0.00
2.80 2.20 2.20 2.35 2.45 3.00 85.00
0 15 30 45 60 75

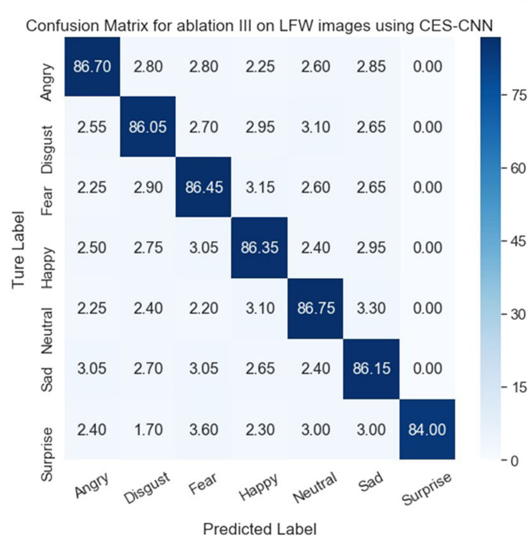
(b)


Confusion Matrix for ablation III on BU-3DFE images using CES-CNN
Ture Label
Predicted Label
Angry Disgust Fear Happy Neutral Sad Surprise
87.55 2.05 2.70 2.60 2.50 2.60 0.00
2.20 87.45 2.35 3.30 2.90 1.80 0.00
2.20 2.55 87.35 2.90 2.55 2.45 0.00
2.75 2.40 2.50 87.45 2.25 2.65 0.00
2.50 2.55 2.60 2.30 87.40 2.65 0.00
2.90 1.85 2.40 2.95 2.40 87.50 0.00
2.90 2.25 2.55 2.30 2.40 2.60 85.00
0 15 30 45 60 75

(c)


Confusion Matrix for ablation III on KDEF images using CES-CNN
Ture Label
Predicted Label
Angry Disgust Fear Happy Neutral Sad Surprise
87.40 2.85 2.70 2.70 1.75 2.60 0.00
2.60 88.05 2.60 2.55 2.45 1.75 0.00
2.50 2.60 87.45 2.20 2.65 2.60 0.00
2.85 2.30 2.90 87.55 2.25 2.15 0.00
2.70 2.35 2.75 2.40 87.15 2.65 0.00
2.60 2.45 2.50 2.75 2.10 87.60 0.00
2.20 2.40 3.05 2.40 2.75 2.20 85.00
0 15 30 45 60 75

(d)


Confusion Matrix for ablation III on LFW images using CES-CNN
Ture Label
Predicted Label
Angry Disgust Fear Happy Neutral Sad Surprise
86.70 2.80 2.80 2.25 2.60 2.85 0.00
2.55 86.05 2.70 2.95 3.10 2.65 0.00
2.25 2.90 86.45 3.15 2.60 2.65 0.00
2.50 2.75 3.05 86.35 2.40 2.95 0.00
2.25 2.40 2.20 3.10 86.75 3.30 0.00
3.05 2.70 3.05 2.65 2.40 86.15 0.00
2.40 1.70 3.60 2.30 3.00 3.00 84.00
0 15 30 45 60 75

(e)

◀ **Fig. 23** Performance evaluation of CES-CNN using confusion matrix for ablation III on **a** JAFFE-O dataset **b** CK+-O dataset **c** BU-3DFE-O **d** KDEF-O dataset **e** LFW-O dataset

Table 4 Evaluation metrics on cross-database validation using CES-CNN

Parameter	Dataset					
	JAFFE	CK+	KDEF	BU-3DFE	LFW	GENKI
Accuracy	93.49	95.76	92.86	94.87	90.66	96.66
Precision	93.58	95.80	92.97	94.93	90.83	96.42
Recall	93.49	95.76	92.86	94.87	90.66	96.86
F1-Score	93.51	95.77	92.88	94.88	90.69	96.45

Table 5 CES-CNN ensemble model accuracy comparison with existing models on JAFFE dataset

Reference	Accuracy (%)
Wen et al. (2017)	50.70
Li et al. (2019b)	51.17
Poux et al. (2021)	91.60
Lopes et al. (2017)	82.10
Minaee et al. (2021)	92.80
Xie and Hu (2018)	94.75
Proposed CES-CNN	**95.39**

Bold represents the better accuracy value of the proposed model

Table 6 CES-CNN ensemble model accuracy comparison with existing models on CK+ dataset

Reference	Accuracy (%)
Xie and Hu (2018)	93.46
Wen et al. (2017)	76.05
Li et al. (2019b)	78.47
Li et al. (2018a)	91.68
Zia et al. (2018)	92.00
Sun et al. (2019)	96.15
Poux et al. (2021)	91.30
Dong et al. (2019)	93.00
Dapogny et al. (2018)	94.30
Proposed CES-CNN	**96.64**

Bold represents the better accuracy value of the proposed model

Table 7 CES-CNN ensemble model accuracy comparison with existing models on KDEF dataset

Reference	Accuracy (%)
Dong et al. (2019)	94.56
Liu et al. (2021)	91.30
Kim et al. (2018)	89.52
Nazir et al. (2018)	93.39
Akhand et al. (2021)	96.51
Proposed CES-CNN	**94.42**

Bold represents the better accuracy value of the proposed model

Table 8 CES-CNN ensemble model accuracy comparison with existing models on BU-3DFE dataset

Reference	Accuracy (%)
Zhang et al. (2016)	79.63
Jie and Yongsheng (2020)	88.70
Wang et al. (2020)	94.18
Shao and Cheng (2021)	78.80
Liu et al. (2021)	84.80
Liu et al. (2018)	94.09
Proposed CES-CNN	**96.24**

Bold represents the better accuracy value of the proposed model

Table 9 CES-CNN ensemble model accuracy comparison with existing models on LFW dataset

Reference	Accuracy (%)
Liu et al. (2018)	85.24
Proposed CES-CNN	**91.33**

Bold represents the better accuracy value of the proposed model

Table 10 CES-CNN ensemble model accuracy comparison with existing models on GENKI dataset

Reference	Accuracy (%)
Kim et al. (2016)	95.25
Chen et al. (2019)	95.55
Umer et al. (2022)	94.67
Proposed CES-CNN	**98.56**

Bold represents the better accuracy value of the proposed model

Conclusion

FER is one of the most challenging research topics in computer vision and is a long-standing problem. Unlike the traditional computer vision problems, this article proposes an effective framework for facial emotion recognition

Table 11 Comparison of classification accuracy on JAFFE-O with existing models

Reference	Accuracy (%) on non-occluded faces	Accuracy (%) on occluded faces
Wen et al. (2017)	50.70	–
Li et al. (2019b)	51.17	–
Liu et al. (2018)	98.13	63.00
Zia et al. (2018)	96.17	–
Poux et al. (2021)	91.60	88.40
Proposed CES-CNN	95.39	93.49

Bold represents the better accuracy value of the proposed model

Table 12 Comparison of classification accuracy on CK+-O with existing models

Reference	Accuracy (%) on non-occluded faces	Accuracy (%) on occluded faces
Liu et al. (2018)	99.02	63.00
Wen et al. (2017)	76.05	–
Li et al. (2019b)	78.47	–
Zia et al. (2018)	92.00	–
Sun et al. (2019)	96.15	–
Poux et al. (2021)	91.3	88.8
Li et al. (2018a)	97.03	93.92
Dong et al. (2019)	93.00	90.30
Dapogny et al. (2018)	94.30	91.50
Proposed CES-CNN	96.64	**95.76**

Bold represents the better accuracy value of the proposed model

Table 13 Comparison of classification accuracy on KDEF-O with existing models

Reference	Accuracy (%) on non-occluded faces	Accuracy (%) on occluded faces
Dong et al. (2019)	94.56	91.42
Liu et al. (2021)	91.30	–
Proposed CES-CNN	94.42	**92.86**

Bold represents the better accuracy value of the proposed model

Table 14 Comparison of classification accuracy on BU-3DFE-O with existing models

Reference	Accuracy (%) on non-occluded faces	Accuracy (%) on occluded faces
Liu et al. (2021)	84.80	–
Liu et al. (2018)	94.09	87.80
Proposed CES-CNN	**96.24**	**94.87**

Bold represents the better accuracy value of the proposed model

Table 15 Comparison of classification accuracy on LFW-O with existing models

Reference	Accuracy (%) on non-occluded faces	Accuracy (%) on occluded faces
Liu et al. (2018)	85.24	64.85
Proposed CES-CNN	**91.33**	**90.66**

Bold represents the better accuracy value of the proposed model

which does not require any feature extraction process. Unlike traditional deep neural networks, the whole face image is not used in this model as input to the CNN. This article tries to exploit the advantages of individual modalities for classifying emotion. As all modalities are used in the training each by a separate CNN, a noisy or a missing modality in the unknown sample would be correctly predicted. The proposed model is rigorously tested on benchmark datasets. Illumination, lightning, orientation, random noise, and scale invariance issues are also addressed in this paper. Experimental results showed that the proposed model outperforms most of the state-of-the-art models.

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Code availability Custom code developed.

Declarations

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Author Contributions Sivaiah Bellamkonda: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing, Visualization. N.P. Gopalan: Conceptualization, Methodology, Writing – review & editing, Supervision, Resources. C. Mala: Conceptualization, Methodology, Writing – review & editing, Supervision, Resources. Lavanya Settipalli: Methodology, Writing – review & editing.

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